The sharp source: a global upper bound and its remaining maxima

Exact original coefficients, all closed parameters

$$N^{-1}\log|a_N(q;\delta)| \quad \le \quad H_\delta(q) \;+\; o(1)$$

Both parities, all row and degree boundaries, sampled zeros and the cap.
This is a uniform upper bound, not a nonzero coefficient asymptotic.

One original shared flow fiber

Port supply 1; ordinary face target 1; protected targets 0.

$$M_D \quad = \quad \max_{q,\boldsymbol{\delta}} \sum_e H_{\delta_e}(q_e), \qquad |S_N| \quad \le \quad e^{N(M_D+o(1))}.$$

At a global maximum on the certified full fiber

Every ordinary incidence is positive; no diagonal port remains.
The original degree constraints give finite common weights $L_f \;>\; 0$.

Protected zeros remain exact; no independent local maxima are assumed.

Complete CROSS

$\delta \;=\; 0$
Both middle loads $< \frac{1}{2}$

No half-seam remains

Strict Green

$0 \;<\; \delta \;<\; \frac{1}{2}$
Strict pole straddling

$$c - b \;=\; \frac{x+y}{\sin^2(\pi\delta)}$$

RIGHT is reflected

Marked cap corner

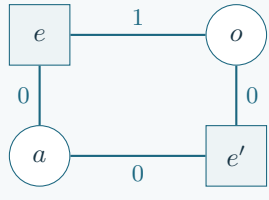
$\delta = \frac{1}{2}, \quad x = b = 0$
$c = y, \quad 0 < \nu < \frac{1}{2}$

At most two; reflected cases included

Every non-CROSS port forces an odd MAIN cycle

The opposite triangles lie in different original ports.
Face demand and common weights rule out matching signs.

Only an all-complete maximum can be phase coherent.



OPEN: an absolute source bound at the volume rate

An odd cycle is a sign obstruction, not a quantitative volume gap.

Regular-window loss must beat the positive envelope.
A flat cap can retain a polynomial alternating remainder.

No general $M_D \;\le\; \Phi_D$ or full filling-period theorem follows here.

Restricted non-CROSS maxima require value at least Φ_D and a strict non-CROSS margin.
The four-cycle is shown in a row/face gauge; its odd sum is invariant.